

SVCS Compression Mode Benchmark

All 4 compression modes verified against the Hawaii H-1 Waimalu highway test clip | 2026-10-01

This report benchmarks all four SVCS compression modes against a single source clip, documents a visual-behavior bug found and fixed in Mode 0 during this pass, and gives before/after frames for each mode so behavior can be checked against spec at a glance.

Source clip	hawaii_H1_waimalu_rush.mp4
Source size / duration	30.04 MB, 120.02s, 3597 frames, 720x480 @ 29.97fps
Camera	Hawaii DOT public live traffic camera - H-1 at Waimalu Exit (Eastbound)
Test settings	segment_seconds=130 (one segment/mode), warmup_frames=60, bg_method=MOG2, object_filter=True, codec/CRF/background_crf left unset (each mode's own default)

What changed in this pass

1. Mode 0 no longer alters the picture

Mode 0 (Live Surveillance) used to run an 8x downscale/upscale blur across the entire background of every frame, independent of CRF - this was the project's original dual-CRF design and the source of its earlier "16.6x / PSNR 41.2dB" headline result, but it meant Mode 0 visibly altered footage rather than just compressing it. Per direction, Mode 0 is now a plain, single-CRF encode of the untouched frame: every frame is recorded, nothing is blurred, blacked out, or recomposited. The code path that triggered the blur (`compress_background` in `src/pipeline/pipeline.py`) is now permanently off for Mode 0.

2. A second, compounding bug in the same code path

Removing the blur alone was not enough. A segment's overall encode quality was being decided once, from whether the single first frame after warmup happened to contain a detection - if that one frame was empty, the entire segment silently fell back to a much heavier CRF (40 by default) for its full duration, even with dense traffic a second later. This made Mode 0's output size effectively random run to run. Mode 0 now always encodes at its own single, predictable foreground CRF (18 by default) regardless of what that first frame contains.

3. Selecting a mode card now resets every option to that mode's defaults

Clicking one of the four mode cards (Live Surveillance / Event Recording / Smart Compress / Object Only) only ever updated the hidden mode selector - CRF, codec, background CRF, segment length, and the object-filter toggle all kept whatever a previously applied preset had left behind (the dashboard applies "Continuous CCTV", a Mode 2 preset, by default on page load). Picking "Object Only" from the cards could silently keep Continuous CCTV's Mode 2 background CRF instead of Mode 3's own settings. Clicking a mode card now clears any active named preset and resets CRF, codec, background-subtraction method, and the object-filter toggle to that mode's plain defaults; the segment-length slider is also fixed so it visually moves when a preset changes it instead of only updating a hidden value.

Auto-Compress already defaulted to Mode 0 before this pass (`DEFAULT_MODE = "mode0"` in `src/gui/services/autocompress_runner.py`) - no change was needed there.

Mode reference (verified against code)

Mode	Frames kept	Background	Typical use
0 - Live Surveillance	All frames, nothing dropped	Untouched - full detail everywhere (as of this fix)	Evidence / archival
1 - Event Recording	Only frames with a detection	N/A - recorded frames are full, untouched frames	Mostly-idle cameras
2 - Smart Compress	Only frames with a detection	A cached "clean" background snapshot, refreshed only after a quiet stretch (~1-1.5s with nothing detected) - stale, not solid black	Busy but not 24/7-critical scenes
3 - Object Only	Only frames with a detection	Always solid black - only the detected boxes show real pixels	Smallest files, lowest context

Important finding specific to this test clip: because the Waimalu footage has near-continuous rush-hour traffic, Mode 2 never got a quiet stretch long enough to capture a real "clean" background - so on this clip, Mode 2's background stayed at its black initial state the entire run and its before/after frame below looks like Mode 3. This is not a bug: it's the documented behavior working as designed, and it means Mode 2 only shows its intended stale-but-real background on cameras that get occasional gaps in traffic.

Benchmark results

Mode	Output size	Ratio vs. 30.04 MB source	Encode time
0 - Live Surveillance	27.23 MB	1.1x	205.6s
1 - Event Recording	24.48 MB	1.23x	207.1s
2 - Smart Compress	8.52 MB	3.53x	208.7s
3 - Object Only	2.72 MB	11.06x	209.5s

Mode 0's ratio (1.1x) is now intentionally close to 1:1 - a predictable, visually-unchanged CRF-18 encode, as it should be for a "just compress it" mode. Mode 1 stays near 1:1 for the same reason on busy footage (nearly every frame qualifies for recording). Mode 2 and Mode 3 compress much harder because they drop idle frames and reduce background detail by design.

Before / after frames (same timestamp, 00:01:00)

Each frame below is pulled from the same one-second mark of its respective output, so the traffic scene is directly comparable across modes.



Source (original, uncompressed)



Mode 0 - Live Surveillance (post-fix: untouched, just compressed)



Mode 1 - Event Recording (full frame, as expected)



Mode 2 - Smart Compress (background stayed black - no quiet moment on this clip)



Mode 3 - Object Only (background always black, by design)

Note the Mode 0 frame above: background is sharp and unaltered end to end, matching the source - confirming the fix. Mode 1 is likewise a full, untouched frame. Mode 2 and Mode 3 both show only the two detected vehicles against black, for the reason explained on the previous page.